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Analysis of Z-Score and total score of athleticism on drafted and undrafted players from the NFL scouting combine

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This study aims to evaluate the effectiveness of z-score and Total Score Athleticism (TSA) in distinguishing between drafted and undrafted NFL players, with a focus on different positional groups. We analyzed NFL Scouting Combine data (2000–2024) using z-score and TSA calculated for both the entire sample and specific positional groups (Skill player, big skill player and linemen), with a final sample of 3,446 players included after filtering. Examine the differences between drafted and undrafted players, as well as the differences among the three positional groups. The classification accuracy was evaluated by ROC analysis and binary logistic regression. Significant differences were observed in z-scores ($P < 0.001$) and TSA ($P < 0.001$) between drafted and undrafted players under the two calculation methods across all positions, with drafted players generally outperforming undrafted ones. ROC analysis indicated that all z-scores on NFL SC demonstrate good performance (AUC > 0.6, $P < 0.001$). Z-score for the 40-yard dash provided the highest AUC (0.637–0.719), while the bench press lowest (0.556–0.631). Position-specific z-score and TSA calculations showed better discriminatory performance compared to the calculation based on the overall sample. This study confirms that both z-scores and TSA are effective in differentiating drafted from undrafted players, with position-specific calculations offering enhanced discriminatory power. The results underscore the importance of tailoring athletic assessments to specific positional needs to improve talent identification. Coaches can understand how z-scores and TSA can be used on National Football League Scouting Combine, and these metrics perform across various positions.

Keywords NFL, American football, Draft, Z-score, Total score of athleticism

Physical fitness testing is a critical method for evaluating athletic ability, assessing training effectiveness, and obtaining feedback on training[1]. The assessments for athletes enable coaches to tailor and adjust training programs to meet the specific needs of their athletes. Similarly, conducting a range of physical performance tests aids in talent identification[2], which has become a standard practice in professional sports and international competitions. The National football league (NFL) is one of the highest level team sport competitions. Several comparative studies have shown that competing in the NFL requires higher physical performance scores, surpassing the demands of other levels of competition[3–5]. The NFL Scouting Combine (SC) is an annual event designed to identify talent for the 32 professional football teams making up the NFL and the testing battery of NFL SC has been going on for decades and its effectiveness has been proven[6]. In that situation, the SC performance have been reported can discriminate drafted and undrafted players[7,8], and is modest predictors of their future performance[6]. For example, broad jump and 40-yd dash display an athlete's lower-body power, are associated with key performance indicators like pass rush defense, pressures, sacks, hits, and hurries[9]. Additionally, it was found that the test scores of players in different positions exhibited distinct characteristics[7,10]. In brief, the NFL SC test battery has some value for the coaches, scouts and teams.

To better utilize the NFL SC, some researchers have studied how to interpret test scores. Robbins et al.[11] analyzed the normalization of data, i.e., ratio-scaled(outcome/BM), and allometrically scaled (outcome/BMa), reported that normalized data provided no advantage over raw data in terms of predicting draft results. Considering there is a different correlation between weight and various test indicators of NFL SC, Robbins et al. suggested explosive movement performance in diverse populations warrant normalization in a subsequent

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study. According to Nuzzo et al.[12], it is more appropriate to use allometric scaling with derived allometric parameters and they give normative reference values for percentile ranking. Gillen et al.[13]further utilized an allometric scaling model to evaluate player performance, explaining the influence of on physical performance of players in different positions. However, it must be considered that defenders are lighter in weight and scaling the results down to weight can create some problems.

Benchmarking has been a critical aspect for the interpretation of testing data[1], which attracts researchers have conducted a lot of research on "standards". However, no research has yet examined the differences in standardized score (z-score) among players in different positions and between drafted and undrafted players. In data analysis and interpretation, z-score and raw score serve different roles and have distinct advantages. Z-score are calculated by the mean and standard deviation from the test score: (test score -mean)/standard deviation, which has been widely used in the research fields of medicine and public health[14,15]have certain advantages in cross-sectional studies. Z-score has been used in some sports science studies[16], but they have not been replicated in studies of physical performance[17]especially in the data analysis of NFL SC. In some moments, such as when selecting athletes, coaches may be less focused on the raw scores of individual athletes and more interested in how those scores rank among teammates[18]. Standardizing physical fitness test scores overcomes the limitation of the raw score, can more effectively categorize different levels. Furthermore, physical fitness assessment needs to focus on general athleticism rather than a particular capacity. Total score of athleticism, which was integrated by Turner et al.[18]provide a holistic perspective to judge the physical performance. The TSA was provided by summing Z-scores for all tests together have been used in previous studies to estimate general physical performance[19]. To sum up, individuals can be compared horizontally through the Z-score. TSA can also conduct quantitative analysis from an overall perspective. However, no one has studied the Z-scores and TSA of NFL SC athletes. Therefore, the aim of this study is to examine the differences in Z-score and TSA of drafted and undrafted player in NFL Scouting Combine, and analysis the position differences. We hypothesize that z-scores of physical fitness tests on NFL SC are associated with draft results, drafted players may have higher athleticism score, and different position players exhibit different characteristic.

Methods

Experimental approach to the problem

Z-score $= x - \mu/\sigma$, Where x is the individual testing value, μ is the mean, and σ is the standard deviation. TSA is the sum of the Z-scores of all tests. To determine whether z-score and TSA on NFL Scouting Combine test results can distinguish draft results, testing data and draft results were retrieved from Open data platform Pro Football Reference (pro-football-reference.com) which have been used in other studies[9].

Firstly, calculate the z-score of each athlete for each test within the overall sample and sum these scores to obtain the TSA. Secondly, for further analysis of differences across positions, treat players from different positions as separate samples, calculate the z-score and TSA for each athlete within their respective sample.

Subjects

The sample used for this study included players who participated in the NFL Scouting Combine between the years of 2000 and 2024. Data for 8,368 subjects were downloaded. Players who did not complete all of the tests were excluded from the study, and these positions including long snappers, kickers, punters and Quarterback were also excluded because the specificity of evaluations and the small sample size. 3446 subjects were finally included in the study, and all data used in the study were grouped into skill players (SP), big skill players (BSP), and linemen (LM) based on previous studies[8,20]. The general information of drafted players are shown in Table 1. Informed consent was not required since all data were publicly available, and no identifiable information was disclosed.

Procedures

This was a secondary data analysis study and the NFL Scouting Combine test procedures have been previously reported[9,21]. Data from 40-yd sprint, vertical jump, bench press, broad jump, 3 cone drill and shuttle run were analyzed:

40-yd Sprint. Standing behind the starting line from a 3-point stance, players sprint for 40-yd (approximately 36.6 m) at maximum speed.

	Mean (± SD)
Height (m)	187.81(6.89)
Body mass (kg)	113.61(21.22)
40-yd sprint (s)	4.80(0.31)
Vertical jump (cm)	83.07(10.83)
Bench press (reps)	20.87(6.39)
Broad jump (cm)	288.98(24.20)
3 Cone drill (s)	7.30(0.42)
Shuttle run (s)	4.41(0.26)

Table 1. Descriptive statistics for NFL scouting combine athletes($N = 3446$).

Vertical Jump. From a stable two-footed stance, players execute a countermovement jump, using arm swing to reach the highest possible vane. Any foot movement before takeoff results in a failed attempt. Jump height is determined by the difference between the highest vane reached and their initial standing reach, measured to the nearest 0.01 m. All athletes performed 2 attempt and the best was recorded.

Bench press. Players perform a maximum number of repetitions of bench press using a weight of 220 lbs (approximately 102.1 kg). For each repetition to count, the bar must be lowered to touch the chest, briefly paused, and then returned to full arm extension at the starting position.

Broad Jump. Standing behind the jumping line, players jump forward as far as possible with arm swing and countermovement. The distance is measured from the jumping line to the heel of closest landing foot. All players performed 2 attempts of broad jump and the longest is recorded.

3 Cone Drill. Three cones are arranged in an "L" shape, with a 5-yard (approximately 4.6 m) distance between each cone. The athlete starts at the first cone with a 3 point stance, sprints to the second cone(the apex of the "L" shape), and then returns. The player then sprints again to the second cone, changes direction towards the third cone, and finally runs back to the second cone and returns to the first cone to complete the test.

Shuttle Run. The player sprints forward 5 yards, quickly reverses direction to sprint back 10 yards, then changes direction again to sprint forward 5 yards, returning to the starting line to complete the test.

Statistical analyses

All data were collected using Microsoft Excel 2021 and are presented as means and SD. IBM SPSS Version 27 (SSPS Inc., Chicago, IL, USA) were used to process statistical analyses. All graphical representations and data visualizations were created using Origin Pro 2024 (OriginLab Corporation, Northampton, MA, USA).

Z-scores and TSA for each test were calculated in the entire sample, and perform separate calculations at different positions. The Shapiro–Wilks test was used to assess the normal distribution. Differential analysis was conducted on z-score and TSA between the drafted and undrafted players. For normally distributed data, independent t-tests were performed for drafted and undrafted players. And the analysis of variance (ANOVA) was employed to examine the differences among different position groups. For skew distribution data, Mann-Whitney U test was conducted on drafted and undrafted players. And the Kruskal-Wallis H test was employed to examine the differences among different position groups. The effect sizes of the independent t-tests and ANOVA were Cohen's d and η[22]. Receiver operating characteristic curve (ROC) is an effective tool for evaluating classification models that has been widely applied in fields such as biomedicine and clinical research[23–25]. The ROC were calculated to determine the optimal cutpoints for z-score and TSA for distinguishing draft results. Simultaneously calculating the area under curve (AUC) to quantify the performance of the classification model; When AUC is between 0.5 and 0.7, there is lower accuracy; When the AUC is between 0.7 and 0.9, there is a certain degree of accuracy; When AUC > 0.9, there is high accuracy[26]. A binary logistic regression was performed according to three position groups. Spearman correlation analysis was performed to examine the associations among z scores and TSA. The significant level (P value) was set as 0.05.

Results

In the z-scores and TSA calculated based on the entire sample, we compared the differences between players in different positions as well as between selected and non-selected players (Table 2). Among the drafted players, significant positional differences were found in both z-scores ($P = 0.000$) and TSA ($P = 0.000$, $\eta 2 = 0.642$). Similarly, positional differences in z-scores ($P = 0.000$) and TSA ($P < 0.001$, $\eta 2 = 0.639$) were also observed among undrafted players. When draft results are not considered, significant differences in z-scores and TSA are observed among the BP, SP, and LM ($P < 0.001$, $\eta 2 = 0.612$). On the other hand, significant differences on z-scores and TSA were found when comparing drafted and undrafted players(P = = 0.000). Further examination of drafted and undrafted players across different position groups revealed significant differences on z-scores within each position. TSA also showed significant differences in both drafted and undrafted of SP ($P < 0.001$, Cohen's d = 1.94), BSP ($P < 0.001$, Cohen's d = 2.42) and LM ($P < 0.001$, Cohen's d = 3.03).

In the further comparison of z-scores and TSA calculated based on data from different position groups, there are also exist significant differences in all z-scores and TSA ($P = 0.000$) between drafted players and undrafted

	ALL (n = 3446)		Skill players (n = 1318		Big skill players (n = 1067)		Linemen (n = 1061)	
	Drafted (n = 2288)	Undrafted (n = 1158)	Drafted (n = 489)	Undrafted (n = 829)	Drafted (n = 737)	Undrafted (n = 330)	Drafted (n = 722)	Undrafted (n = 339)
TSA	0.49(4.03) *	−0.97(4.27)	3.41(1.91) *	2.01(1.98)	1.77(2.34)	−0.14(2.57)	−4.16(2.94)	−6.09(3.22)
40-yd sprint	0.07(0.98) *	−0.14(1.02)	0.96(0.28) *	0.70(0.33)	0.26(0.42)	−0.06(0.48	−1.13(0.59)	−1.45(0.63)
Vertical jump	0.08(0.98) *	−0.16(1.00)	0.74(0.68) *	0.46(0.70)	0.29(0.72)	−0.07(0.80)	−0.88(0.71)	−1.16(0.74)
Bench press	0.10(1.00) *	−0.21(0.96)	−0.71(0.73) *	−0.85(0.74)	0.31(0.73)	−0.02(0.74)	0.83(0.81)	0.53(0.79)
Broad jump	0.07(0.99) *	−0.15(0.99)	0.80(0.60)	0.54(0.56)	0.30(0.63)	−0.02(0.61)	−0.98(0.70)	−1.28(0.72)
3 Cone drill	0.07(0.96) *	−0.14(1.05)	0.81(0.48) *	0.60(0.51)	0.28(0.57)	0.01(0.63)	−1.00(0.69)	−1.36(0.83)
Shuttle run	0.08(0.97) *	−0.15(1.03)	0.79(0.56) *	0.54(0.57)	0.30(0.60)	0.01(0.63)	−0.97(0.72)	−1.13(0.78)

Table 2. Means (± SD) of z-scores and TSA# on NFL SC tests for drafted and nondrafted players. Note: * means drafted players performed significantly better than undrafted players ($P < 0.001$). # means z-score and TSA were calculated by whole sample.

(Table 3). However, when draft results are not considered, there is no difference on z-scores for 40 yd sprint (P = 0.874), vertical jump (P = 0.874), bench press (P = 0.720), broad jump (P = 0.821), 3 cone drill (P = 0.804) and shuttle run (P = 0.996), and TSA (P = 0.955) among the three group. When considering draft results, drafted players show significant positional differences only in the z-score of 40-yard sprint (P = 0.011), and the results of group comparison showed that the difference was mainly between SP and LM groups (P = 0.008). There is no significant difference in other z-scores for vertical jump (P = 0.676), bench press (P = 0.718), broad jump (P = 0.874), 3 cone drill (P = 0.902) and shuttle run (P = 0.900) and TSA (P = 0.741). Undrafted players show significant positional differences only in the z-score of bench press (P = 0.004), and show no differences in 40 yd sprint (P = 0.1133), vertical jump (P = 0.553), broad jump (P = 0.317), 3 cone drill (P = 0.939), shuttle run (P = 0.768), and TSA (P = 0.318).

Figure 1 illustrated the ROC Analysis of 40 yd sprint, vertical jump, bench press, broad jump, 3 cone drill and shuttle run based on z-scores calculated separately for SP, BSP and LM groups, with ROC curve, area under curve and optimal cut-off value. All z-scores for the NFL SC tests showed significance in the ROC analysis (P < 0.001), with the 40-yard sprint achieving the highest AUC (0.637–0.719) and the bench press having the lowest AUC (0.557–0.613). Figure 2 representing ROC curve, area under curve and optimal cut-off value of TSA based on the z-scores calculated from the whole sample, as well as the ROC curves of TSA after calculating the z-scores separately for the SP, BSP, and LM groups. Both TSA calculation methods showed significance in the ROC analysis (P < 0.001), but the TSA calculated based on the entire sample had a lower AUC (0.604).

Binary Logistic Regression results were shown in Table 4. The results indicate that improvements in z-scores for various tests are associated with a higher probability of athlete selection(P < 0.001). The bivariate correlation analysis is displayed in Fig. 3. The z-scores of 40-yard dash, broad jump, vertical jump, 3-cone drill, and shuttle run are highly correlated among the SP, BSP, and LM groups. However, among players in all three positions, the bench press shows a weaker correlation (r < 0.3) with these indicators.

Discussion

The current study set out to examine the difference and correlation on z-score and TSA between drafted and undrafted players among different positions, who participated in the NFL SC from 2000 to 2024, and analyze the classification performance. The Z-score, which is a standard score for cross-sectional comparisons[15], is derived by dividing the difference between the test score and the average of the entire population by the standard deviation. The TSA is an holistic athlete profiling approach based on Z-score[18]. In this study, two z-scores were calculated, one was the z-scores and TSA in all samples, and the other was the z-scores and TSA in position groups. The primary results of this study consistent with the stated hypothesis, revealed significant differences between drafted and undrafted players among SP, BSP and LM groups. The ROC curve and AUC are shown in Figs. 1 and 2, indicating good predictive accuracy.

In the Z-scores and TSA calculated for the overall sample, significant differences were found between drafted and undrafted players across all z-scores on NFL SC tests. Uniformly, there are significant differences in the z-scores and TSA between drafted and undrafted players in the BP, SP, and LM groups (Table 1). Regardless of players' position, drafted players had significantly superior values of z-scores and TSA on NFL SC compared with undrafted, indicating that drafted players demonstrating greater performance on NFL SC. This finding further validates previous reports[8], that the NFL SC is an effective tool for athlete selection, with players who perform better in this test having a higher likelihood of being drafted[6]. Considering well-developed physical fitness is a crucial factor for team selection in professional American football[27], and the z-score represents a positional ranking of athletes on physical tests and can effectively distinguish their relative athleticism, coaches and practitioners can use Z-scores and TSA to differentiate. On the other hand, previous studies have found different physiological and anthropometric characteristics between different positions[27]. The current study find that SP performs best in z-scores of 40 yd sprint, vertical jump, broad jump, 3cone drill and shuttle, followed by BSP and LM. In the bench press, the opposite is true, with LM performing best, followed by BSP and SP. SP and BSP are more excellent in terms of speed and sensitivity, and LM group has better strength. These results confirm the findings of previous studies that relationships between various sprint and jump abilities warrant positional consideration[10]. In short, the movement patterns and physiological demands of different positions exist differences, understanding theses characteristic could help the development of strength and conditioning

	Skill players (n = 1318) ^		Big skill players (n = 1067)		Linemen (n = 1061)	
	Drafted (n = 489)	Undrafted (n = 829)	Drafted (n = 737)	Undrafted (n = 330)	Drafted (n = 722)	Undrafted (n = 339)
TSA	0.98(3.29) *	−1.66(3.49)	0.93(3.66) *	−2.09(4.03)	0.84(4.03) *	−1.79(4.41)
40-yd sprint	0.28(0.87) *	−0.49(1.01)	0.21(0.91) *	−0.47(1.02)	0.16(0.95) *	−0.34(1.01)
Vertical jump	0.14(0.97) *	−0.25(0.99)	0.14(0.94) *	−0.33(1.04)	0.12(0.97) *	−0.25(1.00)
Bench press	0.69(0.99) *	−0.11(1.00)	0.13(0.97) *	−0.30(0.98)	0.11(0.99) *	−0.25(0.97)
Broad jump	0.15(1.00) *	−0.26(0.93)	0.15(0.98) *	−0.34(0.94)	0.13(0.97) *	−0.28(1.00)
3 Cone drill	0.15(0.95) *	−0.26(1.02)	0.13(0.94) *	−0.31(1.04)	0.14(0.91) *	−0.31(1.09)
Shuttle run	0.16(0.97) *	−0.27(0.99)	0.14(0.96) *	−0.31(1.00)	0.15(0.94) *	−0.33(1.02)

Table 3. Means (± SD) of z-scores and TSA^ on NFL SC tests for drafted and nondrafted players. Note: * means drafted players performed significantly better than undrafted players (P < 0.001). ^ means z-score and TSA were calculated by position groups.

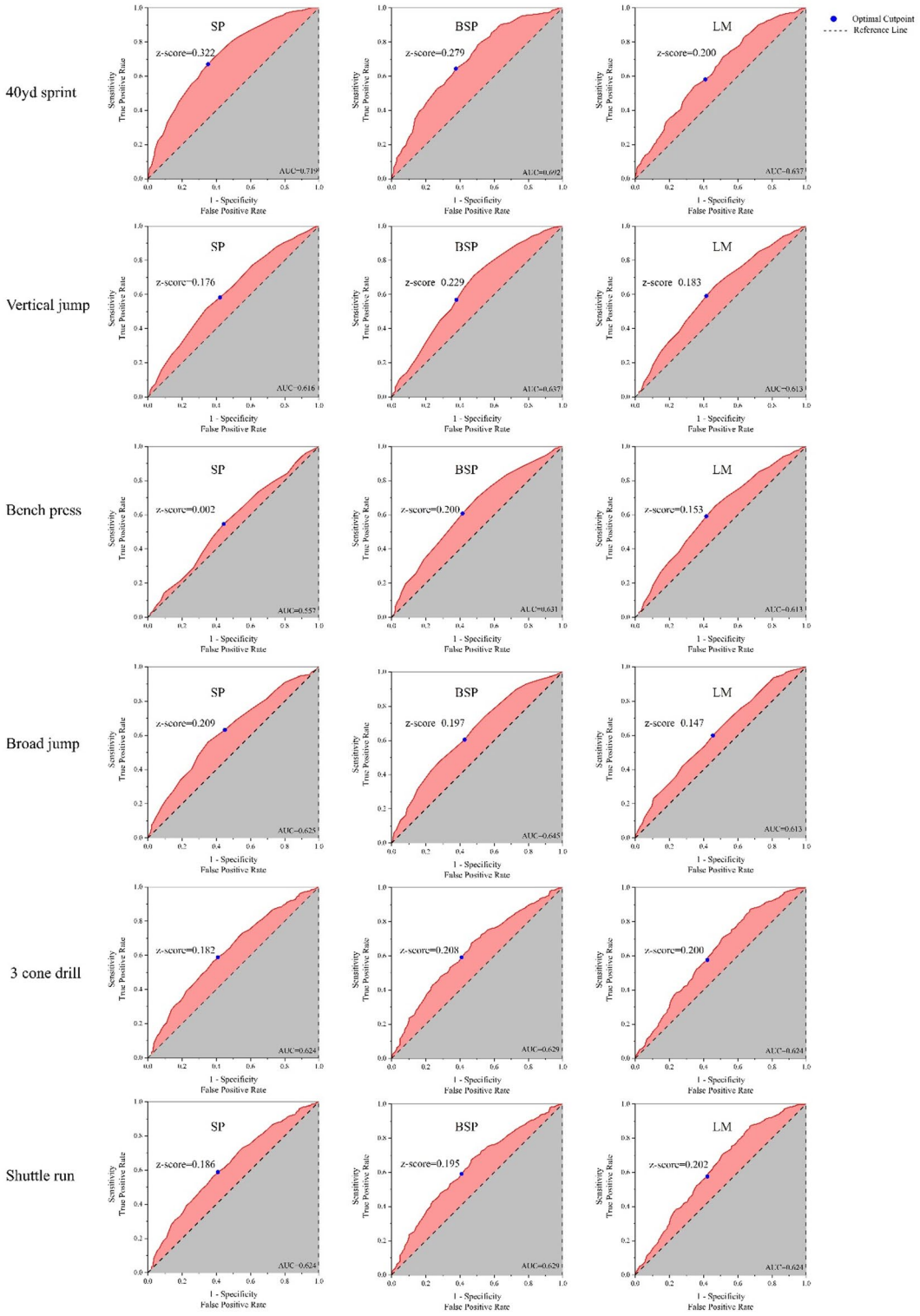

Fig. 1. Results of ROC Analysis for the z-score on NFL Scouting Combine Test. SP means skill player; BSP means big skill player means LM: linemen.

programmes. Strength and conditioning coaches can find the disadvantage of the athlete and strengthen it according to their z-scores and TSA from the overall sample of the athlete participating in the draft.

In the Z-scores and TSA calculated for separate groups, there were significant differences in z-scores and TSA for each test between drafted and undrafted players in each group. This finding further validates previous analyses of differences (based on Z-scores for the overall sample), which indicate that the relative athleticism of drafted players in all positions was better than that of undrafted players. However, the generally non-existent differences

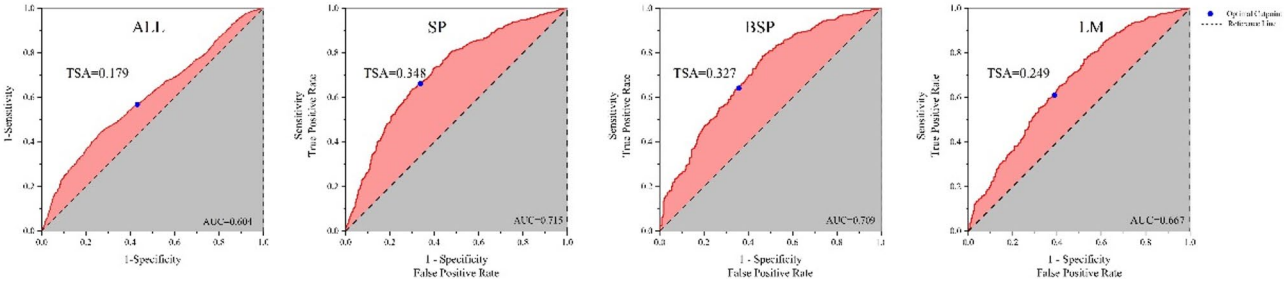

Fig. 2. Results of ROC Analysis for the TSA on NFL Scouting Combine Test. ALL means TSA calculated by z-scores on whole sample; SP means TSA calculated by z-scores on skill players; BSP means TSA calculated by z-scores on big skill player; LM means TSA calculated by z-scores on linemen.

	Position	B	SE	*P*	OR	95%CI
TSA	SP	0.231	0.019	< 0.001	1.260	1.213 ~ 1.308
	BSP	0.209	0.020	< 0.001	1.232	1.185 ~ 1.280
	LM	0.153	0.017	< 0.001	1.165	1.126 ~ 1.206
40-yd sprint	SP	0.899	0.071	< 0.001	2.456	2.137 ~ 2.822
	BSP	0.750	0.076	< 0.001	2.116	1.824 ~ 2.456
	LM	0.539	0.072	< 0.001	1.715	1.489 ~ 1.974
Vertical jump	SP	0.420	0.061	< 0.001	1.522	1.350 ~ 1.715
	BSP	0.514	0.073	< 0.001	1.672	1.450 ~ 1.927
	LM	0.395	0.070	< 0.001	1.484	1.295 ~ 1.702
Bench press	SP	0.190	0.058	0.001	1.209	1.078 ~ 1.355
	BSP	0.485	0.074	< 0.001	1.625	1.406 ~ 1.877
	LM	0.390	0.071	< 0.001	1.477	1.286 ~ 1.696
Broad jump	SP	0.447	0.062	< 0.001	1.563	1.385 ~ 1.764
	BSP	0.552	0.075	< 0.001	1.736	1.500 ~ 2.009
	LM	0.434	0.070	< 0.001	1.544	1.345 ~ 1.771
3 Cone drill	SP	0.442	0.060	< 0.001	1.556	1.382 ~ 1.751
	BSP	0.455	0.069	< 0.001	1.576	1.377 ~ 1.804
	LM	0.482	0.070	< 0.001	1.619	1.413 ~ 1.856
Shuttle run	SP	0.455	0.061	< 0.001	1.576	1.399 ~ 1.775
	BSP	0.470	0.070	< 0.001	1.600	1.394 ~ 1.835
	LM	0.517	0.071	< 0.001	1.677	1.459 ~ 1.927

Table 4. Binary logistic regression Results.

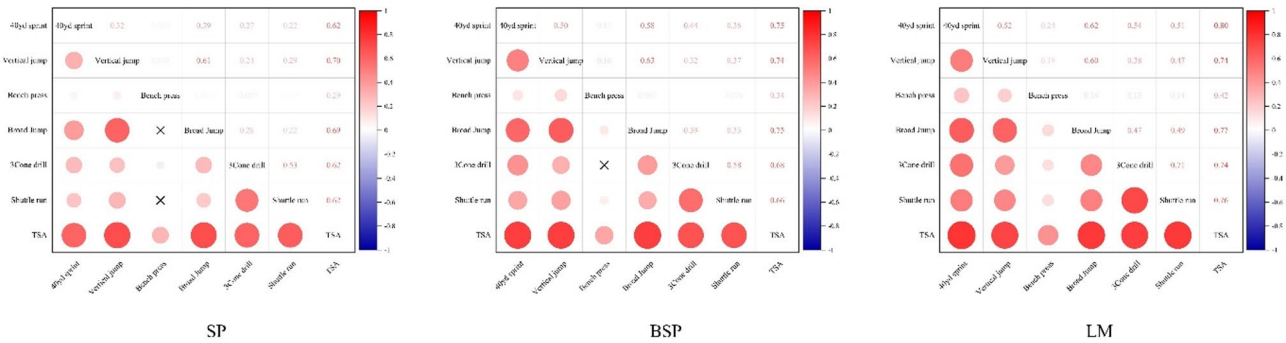

Fig. 3. Correlation Heat Map of z-score and TSA on NFL SC tests. Legend: ✕ means $P > 0.05$.

among different groups in this calculate method may be due to the fact that Z-scores are not calculated in the same sample. Therefore, we did not report these detail differences in our findings. This method of calculation is mainly to allow for subsequent classification tests to be more detailed, rather than to analyze the differences between them because different test results have different significance for different position groups[9].

In order to analyse the classification performance of z-scores for NFL SC tests on drafted and undrafted players, ROC analyses and binary logistic regression were used based on z-scores calculating by SP, BSP and LM groups. The result is shown in Figs. 1 and 2; Table 4. ROC-curves are often used to test overall quality of classification models. The results showed that all tests had good accuracy (most of the AUC > 0.6). Although an AUC theoretically greater than 0.8[26] indicates excellent classification performance. However, in the topics of talent selection and physical fitness testing, the direct prediction of the selection result by a single test index does not conform to the theory and the actual situation. These results indicate that the various test indicators of the NFL SC can distinguish the selected and unselected players in different positions to a certain extent, proving the rationality of the test indicators. According the AUC, regardless of position, in the NFL SC test, z-score of 40 yd was the best at distinguishing drafted and undrafted. The binary logistic regression further confirms this phenomenon. For every 1 point increase in the z-score of 40 yd sprint, the probability of a player being selected increases by 1.7–2.4 times. This finding reveals that the drafted SP, BSP and LM players placed great emphasis on 40 yd sprint ability, consistent with previous studies[8] that elite players typically displayed greater short-distance sprint performance. The high discriminant performance of sprint ability may be determined by the race characteristics. American football can be characterized as a turn-based sport based on high intensity, repetitive sprints. The average duration of a play in American football was about 5 seconds[28]and the sprint ability is high correlated with player performance[9]. The optimal cut-off values at different positions at 40 yd are also different, and the SP value is higher, indicating higher requirements in this respect. The higher sprinting demand of SP may be related to the positioning on the field, where they need to sprint more than other positions[29]. Moreover, z-scores on vertical jump, broad jump, 3cone drill, and shuttle run exhibit similar classification performance and were all able to distinguish whether they were drafted or not. Vertical jump and broad jump are usually considered a kind of power index[27] rather than a specific performance indicator, the classification performance is therefore down compared to the 40 yd sprint. The results of vertical jump (OR = 1.4–1.6) and broad jump (OR = 1.5–1.7) are similar in binary logistic regression. Our research found that different positions have different needs for jumping ability consistent with the findings of Robbins[10]. Additionally, although Change of direction (COD) is a fundamental maneuver in American football[30] and high correlation was found in sprint and agility[30], COD may exist significant differences between 3 cone drill, shuttle run and real competition. Differences in the Angle of change lead to significant changes in biomechanical characteristics[31], indicating that different capabilities are required[32], which also leads to a decline in classification performance compared to sprint. And the results of the optimal cut-off values show that the z-scores of BSP and LM drafted players are higher than those of SP on 3 cone drill and shuttle run. Higher agility requirement for BSP and LM of BSP and LM may be due to the positional differences[33]. BSP and LM have more accelerating, decelerating, and physical collisions[29]. However, bench press has relatively weak performance in distinguishing between drafted and undrafted players, with a probability increase of only 1.2–1.6 times for every 1 point increase in z-score. In particular, SP position athletes have the lowest discrimination performance (AUC = 0.556), and the selection probability of athletes increases 1.2 times with each increment of z-score (OR = 1.209, $P < 0.001$). The classification performance of BSP and SP (BSP: AUC = 0.631, LM: AUC = 0.613) has increased, and they placed more emphasis on upper body strength (OR = 1.477–1.625, $P < 0.001$) than SP. Besides, the optimal cut points for z-score on bench press can demonstrate the difference in demand for different locations. The optimal cu-toff value of SP (0.002) is much lower than that of BSP(0.197) and LM(0.147).

It is worth mentioning that z-scores of SP and BSP on sprint and agility tests were either irrelevant or weakly correlated with the bench press, but the correlation increased in LM, verifies this characteristic (Fig. 3). Namely, LM is more demanding for upper body strength. This difference of upper body strength may relate to the different competition requirements described earlier. In short, although AUC > 0.7 was found in only a small number of tests, this is still a useful reference. As mentioned before, physical fitness is a key factor in American football talent identification and orientation[34], but not the only[35]. High level of performance on NFL SC contributes to effective playing ability. Our findings all show that z-scores based on location differences may provide more valid reference.

Furthermore, added ROC analysis of TSA based on overall sample calculation and grouping, as shown in Fig. 2. A higher TSA indicates that athletes perform relatively better on all tests of the NFL SC. Based on the z-scores calculated from the overall sample, TSA is able to effectively distinguish between selected and omitted players (AUC = 0.604). Consistently, TSA based on Z-scores of grouped samples has better classification performance (SP: AUC = 704; BSP: AUC = 0.719; LM: AUC = 0.667). This further suggests that z-scores calculated based on different locations are more efficient on distinguishing characteristics of players. But it is worth noting that higher TSA does not mean a higher success rate. The results of binary logistic regression showed that the probability of athletes being selected increased by 1.1–1.2 times for every 1 point increase in TSA (OR = 1.165–1.260, $P < 0.001$), which was well below the z-scores on 40-yd sprint and other tests (Table 4). This is consistent with the analysis on z-scores, that players in different positions have different characteristics and advantages in the NFL SC tests. For example, LM perform better in the bench press, but their performance in sprint test is weaker than SP and BSP, and the requirements are relatively low. SP has better sprint ability, but its upper body strength is relatively weak. Therefore, we recommend that the NFL develop more targeted tests based on player positions.

Limitations

Previous studies have not investigated the application of standard score (z-score) on NFL SC. Z-scores, which represent how athletes rank on tests, are a great analytical tool[15], especially in large samples such as the NFL SC over several decades. In this study, we presume z-scores on NFL SC could distinguish drafted and undrafted players and verify the differences between different positions.

However, several limitations should be acknowledged. Firstly, the findings are specific to NFL SC participants and may not generalize to athletes in other leagues. This limitation reduces the applicability of the results to broader athletic populations. Secondly, the grouping of positions (SP, BSP, and LM) simplifies the nuanced differences within each group. More detailed positional classifications might reveal finer distinctions in physical and skill-related performance. On the other hand, some confounding factors such as training background, collegiate performance were not included in the analysis. Future research should consider collecting and integrating training logs, university competition data and detailed anthropometric information to further analyze the true contribution of each performance indicator. Addressing these limitations in future research could improve the predictive power of z-scores and TSA, while providing a more comprehensive understanding of the factors influencing NFL draft outcomes.

Practical applications

In this paper, two application methods of z-scores are proposed and analyzed. According to the results, drafted players at all positions showed better athleticism in NFL SC. The z-scores and TSA (calculated based on the overall sample) of players in different positions also differ. This study validates the results of previous studies from another perspective and shows that Z-scores and TSA are effective tools to distinguish between drafted and undrafted. However, considering that 'predicting' whether an athlete will be selected is not a scientific question, we suggest using the z-score as a key indicator to distinguish athlete performance levels and to set training goals accordingly. Specifically, regarding the measuring and training, considering the different requirement of players the z-scores calculating on position groups may be more effective. Strength and conditioning professionals who aim to identify weaknesses and deficiencies in their athletes should use the z-scores. By contrast, thought TSA was able to reflect the discriminating effectiveness of the NFL SC test battery, we found that a higher TSA, which was determinted by the testing battery, does not increase the probability of an athlete being drafted. Considering the application of Z-scores and TSA needs to be based on effective test battery. Consequently, we recommend that NFL should develop differentiated testing batteries based on position differences.

Conclusions

This study demonstrates that z-scores and the Total Score of Athleticism (TSA) are effective tools for distinguishing between drafted and undrafted players in NFL SC. And the calculation based on the sample of specific position greatly enhances the discriminatory power. Future research should explore more detailed positional classifications and incorporate a broader range of player characteristics to enhance the comprehensiveness of draft evaluations. Overall, this study contributes to a deeper understanding of how physical performance metrics can guide the identification and selection of players in American football.

Data availability

Data was retrieved from Open data platform Pro Football Reference (pro-football-reference.com).

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Declarations

Competing interests
The authors declare no competing interests.

Ethics declarations
This was a secondary data analysis study and the NFL Scouting Combine test procedures have been previously reported. The studies were conducted in accordance with the local legislation and institutional requirements.

Additional information
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